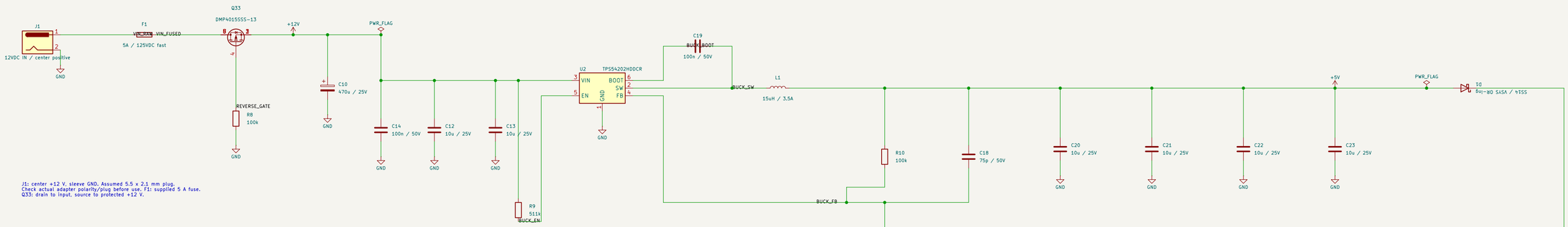
IEEE ARCADE | 30 playfield targets + 2 menu buttons

12 V DC external adapter | Pico 2 W + high-score memory + 32 inputs / 32 lamp outputs

INPUT PROTECTION



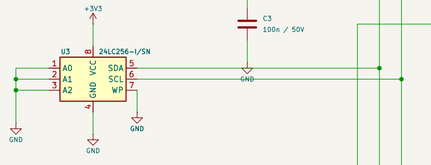
J1: center +12 V, sleeve GND. Assumed 5.5 x 2.1 mm plug.
Check actual adapter polarity/plug before use. F1: supplied 5 A fuse.
Q33: drain to input, source to protected +12 V.

12 V -> 4.95 V BUCK | 2 A DEVICE / 1.8 A DESIGN ALLOCATION

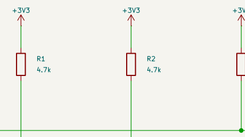
Feedback: 100k / 13.7k -> 4.95 V nominal. C20..23: 4 x 10uF/25 V X7R (~34uF typical at 5 V bias).
Use TI reference layout: short VIN/BOOT/SW loops, clean FB return. Validate load steps/thermal behavior on hardware.

D1 prevents USB backfeed into the buck. VBUS has no external connection.
5 V budget: Pico 0.5 A + logic reserve 0.1 A + display <=1.2 A.
32 lamps x 100 mA allocation + 5 V load at 85% efficiency -> 4.1 A at 12 V.
Lamp/display currents are assumptions; verify actual loads before layout.

CONTROLLER / HIGH-SCORE EEPROM



U3: 0x50. WP low; writes enabled.
Save only changed high scores, at round end.



GPIO14/15: I2C1 at 100 kHz. GPIO13: expander reset.
GPIO0..12: provisional display; spare GPIO reserved for later audio.
Pico: solder castellations, leave antenna keepout clear; USB for programming.
Unused ADC/AGND require no external wiring.

Before layout: verify display protocol/pinout/current, adapter plug/polarity,
and menu-lamp current/terminals. No audio or battery in this revision.
Schematic only: PCB untouched.

BUTTON INPUTS | all 32 pins used as inputs

U1: 0x20
U4: 0x27

MCP23018: all 16 ports support input.
MCP23017 GPA7/GPB7 are output-only; used for lamps here.
Inputs have external pull-ups/RC filters on the button-array sheets.

LAMP OUTPUTS | 32 low-side MOSFET controls

U5: 0x21
U6: 0x22

Firmware: pulse expander RESET low >=10 us.
Write lamp OLAT=0 BEFORE IODIR=outputs. HIGH lights a lamp.
Poll inputs every 1-2 ms; require 10 ms stable state for debounce.

Scoreboard (PROVISIONAL)
File: scoreboard.kicad_sch

Button circuits 1
File: button_bank_1.kicad_sch

Button circuits 2
File: button_bank_2.kicad_sch

Button circuits 3
File: button_bank_3.kicad_sch

Button circuits 4
File: button_bank_4.kicad_sch

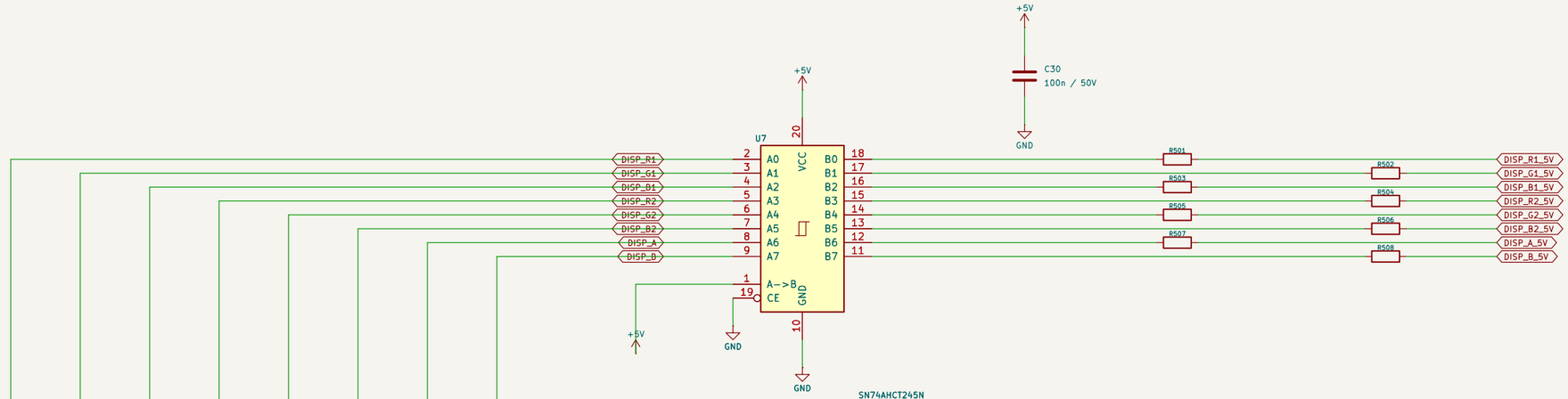
Panel-mounted buttons 1
File: external_buttons_1.kicad_sch

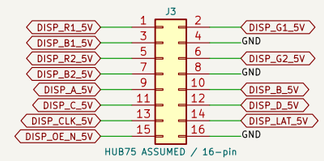
Panel-mounted buttons 2
File: external_buttons_2.kicad_sch

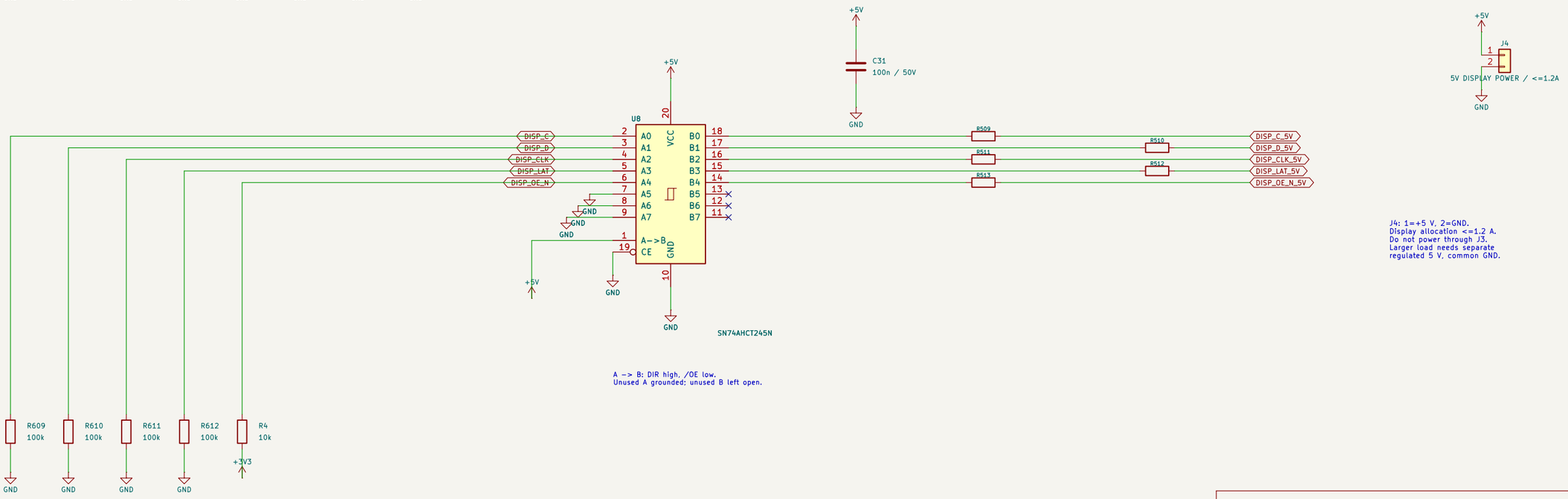
UW-Madison IEEE
Sheet: /
File: IEEEArcade.kicad_sch
Title: IEEE Arcade | Pico 2 W, power, memory and I/O
Size: A2 | Date: 2026-11-05 | Rev: Schematic review
KiCad E.D.A. 10.0.6 | Id: 1/8

EXTERNAL SCOREBOARD | PROVISIONAL HUB75-STYLE, 16 pins

User requested an expected connector; actual panel interface/current NOT confirmed.
P10 specifies pixel pitch, not protocol. This header is NOT I2C. Check panel pinout before use.







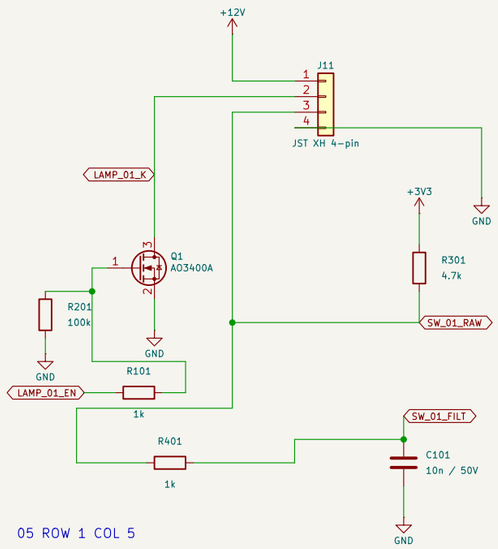
R501..R513: all 33 ohms; values hidden here to keep close signal rows readable.
AHCT accepts 3.3 V inputs; its 5 V outputs drive J3. Pull-downs default signals low; OE_N pull-up blanks at reset.
Start with a short ribbon (<=30 cm); verify signal integrity. Pin8=GND here; some panels use address E.
Confirm panel supply/pin mapping before connection or PCB layout.

UW-Madison IEEE		
Sheet: /Scoreboard (PROVISIONAL)/		
File: scoreboard.kicad_sch		
Title: Provisional 16-pin external scoreboard		
Size: A3	Date: 2026-11-05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 2/8

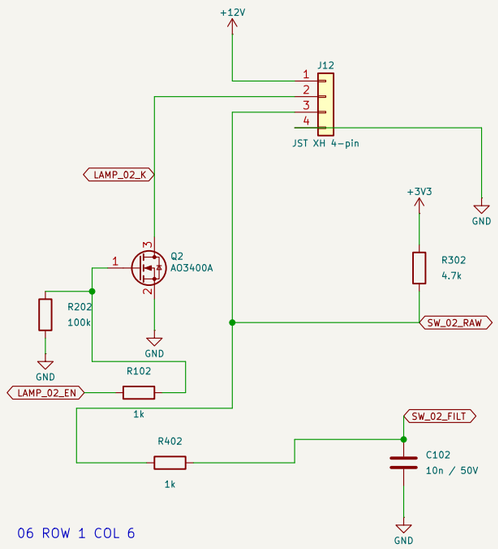
BUTTON HARNESS | bank 1 | four wires per illuminated button

Jx: 1=+12V lamp, 2=switched lamp cathode, 3=switch NO, 4=switch COM/GND.
Adafruit 1194 includes an LED resistor for operation up to 12 V. No extra series resistor fitted.
EG STARTS menu lamps specified as 12 V; current / tab dimensions need confirmation.
Use normally-open contacts; leave any NC switch terminal insulated and unused.

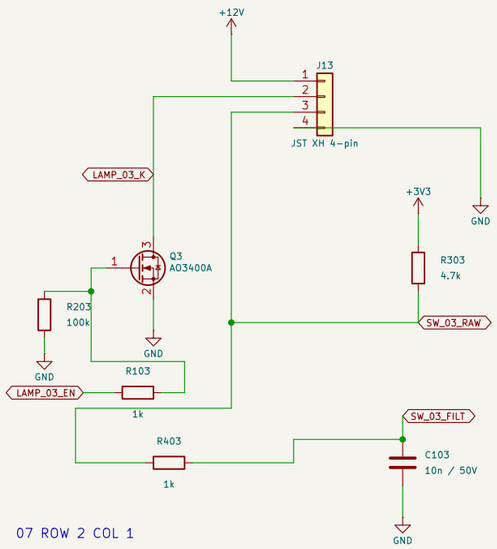
01 ROW 1 COL 1



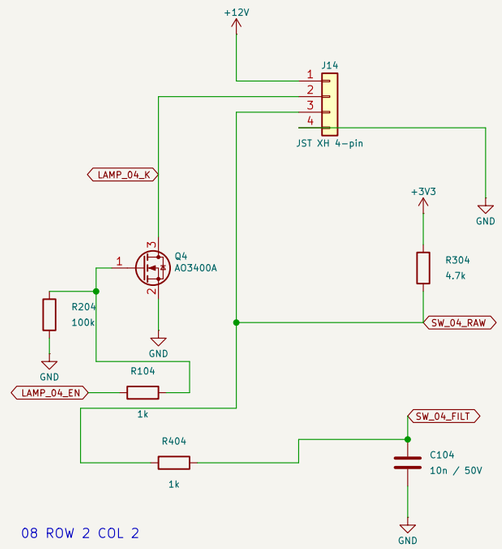
02 ROW 1 COL 2



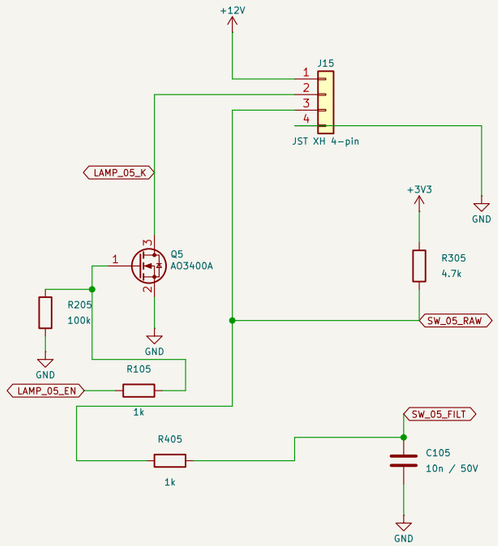
03 ROW 1 COL 3



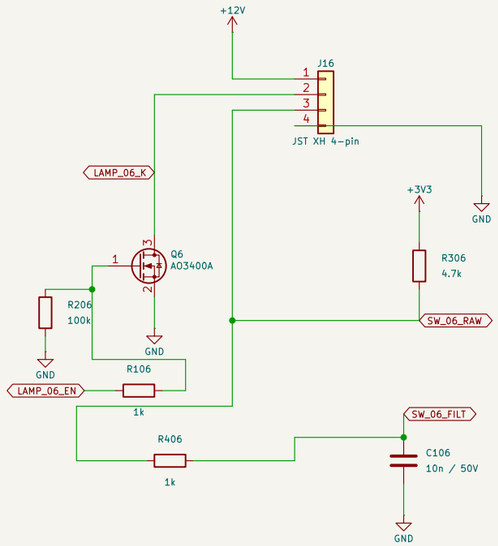
04 ROW 1 COL 4



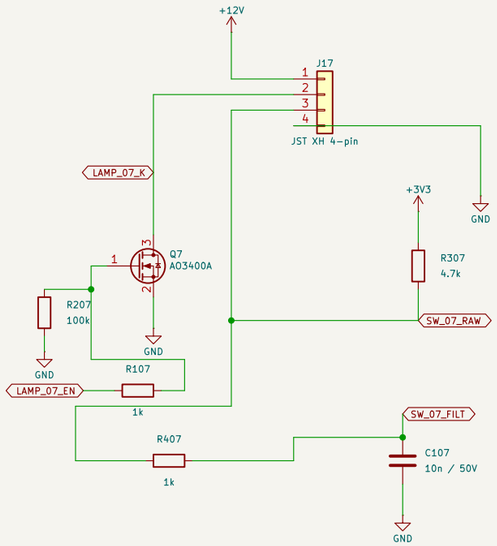
05 ROW 1 COL 5



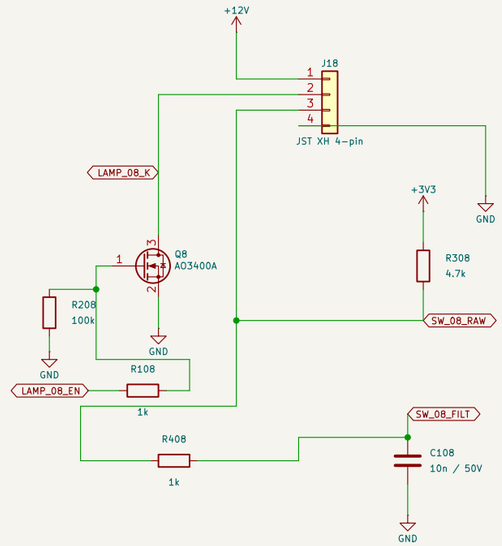
06 ROW 1 COL 6



07 ROW 2 COL 1



08 ROW 2 COL 2



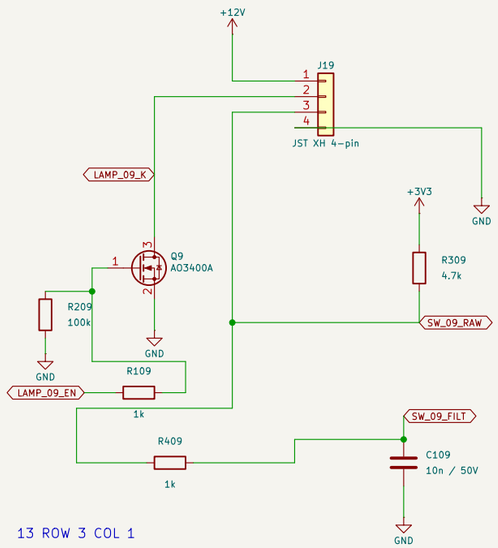
100 mA per lamp is a DESIGN ALLOCATION, not a measured lamp current. 32 lamps = 3.2 A.
XH cable mating parts: XHP-4 + 4 x SXH-001T-P0.6 per button. Accepts 22-28 AWG; strain-relieve.
Adafruit 1194: 3835 pigtails fit switch tabs; 3838 fit lamp tabs. See BOM for pack quantities.
Replace pigtail 2-pin ends with genuine XH contacts / 4-pin housing. Menu-button tab sizes UNVERIFIED.

UW-Madison IEEE	
Sheet: /Button circuits 1/	
File: button_bank_1.kicad_sch	
Title: Button harness / channels 01..08	
Size: A3	Date: 2026-11-05
KiCad E.D.A. 10.0.6	Id: 3/8
	Rev: Schematic review

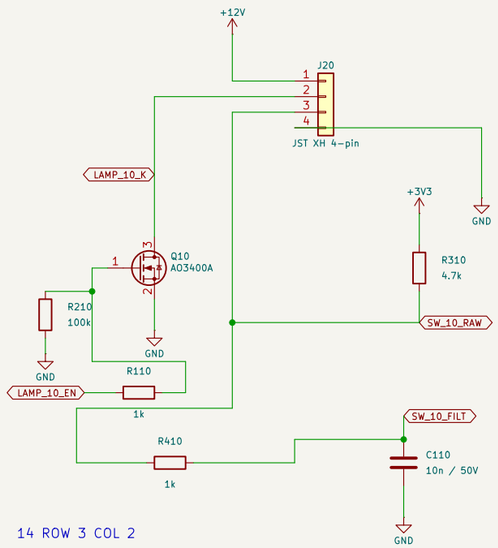
BUTTON HARNESS | bank 2 | four wires per illuminated button

Jx: 1=+12V lamp, 2=switched lamp cathode, 3=switch NO, 4=switch COM/GND.
Adafruit 1194 includes an LED resistor for operation up to 12 V. No extra series resistor fitted.
EG STARTS menu lamps specified as 12 V; current / tab dimensions need confirmation.
Use normally-open contacts; leave any NC switch terminal insulated and unused.

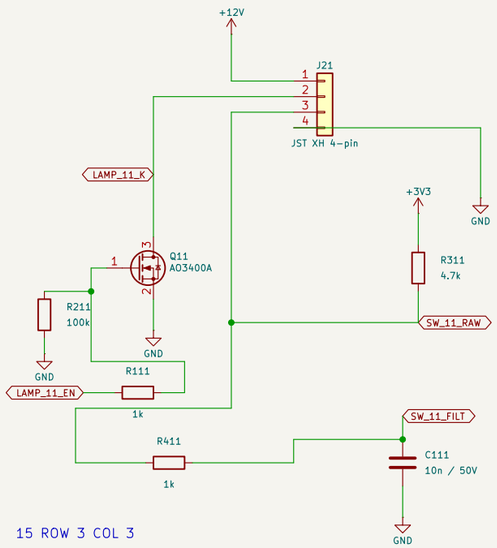
09 ROW 2 COL 3



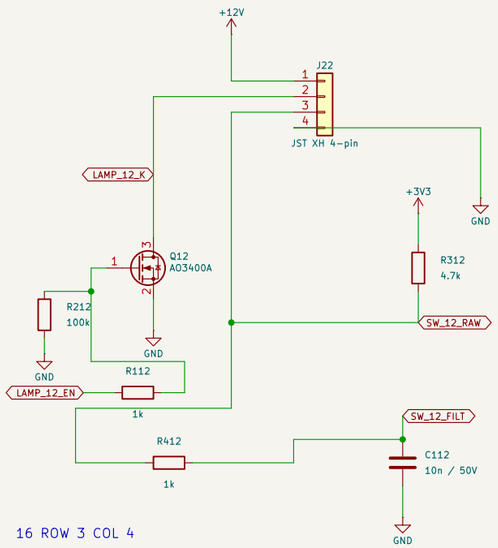
10 ROW 2 COL 4



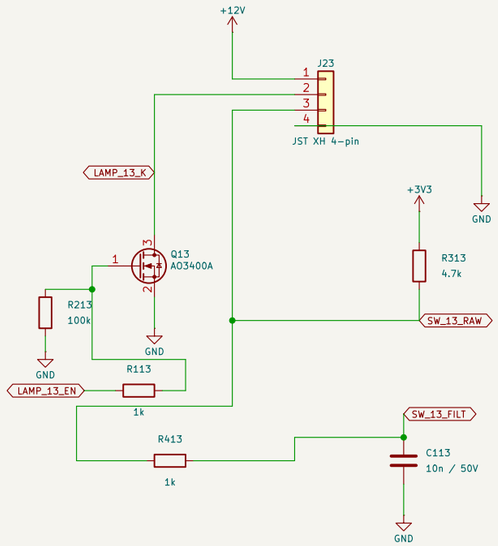
11 ROW 2 COL 5



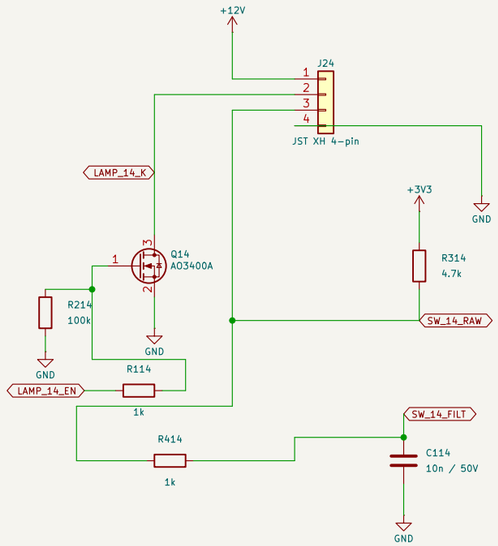
12 ROW 2 COL 6



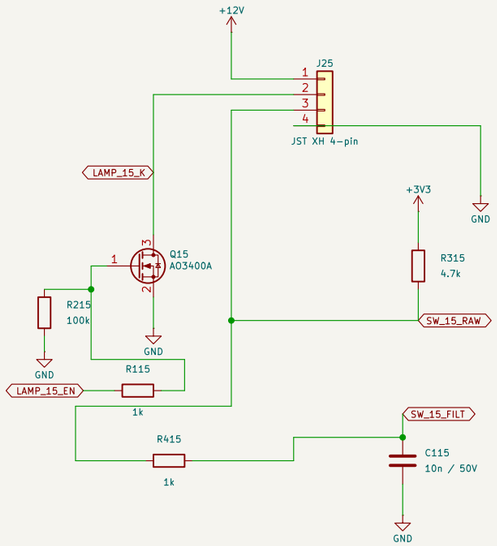
13 ROW 3 COL 1



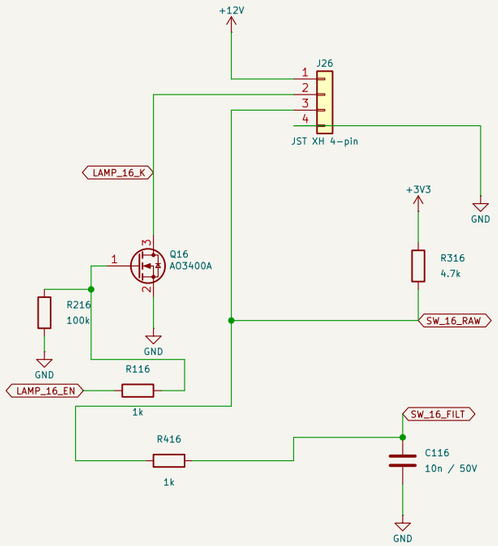
14 ROW 3 COL 2



15 ROW 3 COL 3



16 ROW 3 COL 4



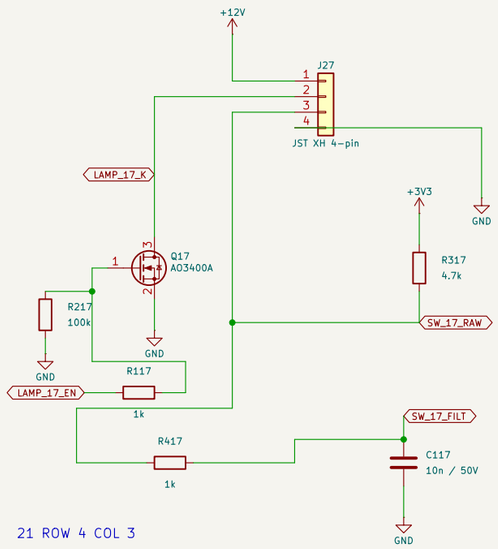
100 mA per lamp is a DESIGN ALLOCATION, not a measured lamp current. 32 lamps = 3.2 A.
XH cable mating parts: XHP-4 + 4 x SXH-001T-P0.6 per button. Accepts 22-28 AWG; strain-relieve.
Adafruit 1194: 3835 pigtails fit switch tabs; 3838 fit lamp tabs. See BOM for pack quantities.
Replace pigtail 2-pin ends with genuine XH contacts / 4-pin housing. Menu-button tab sizes UNVERIFIED.

UW-Madison IEEE
Sheet: /Button circuits 2/
File: button_bank_2.kicad_sch
Title: Button harness / channels 09..16
Size: A3 | Date: 2026-11-05 | **Rev: Schematic review**
KiCad E.D.A. 10.0.6 | Id: 4/8

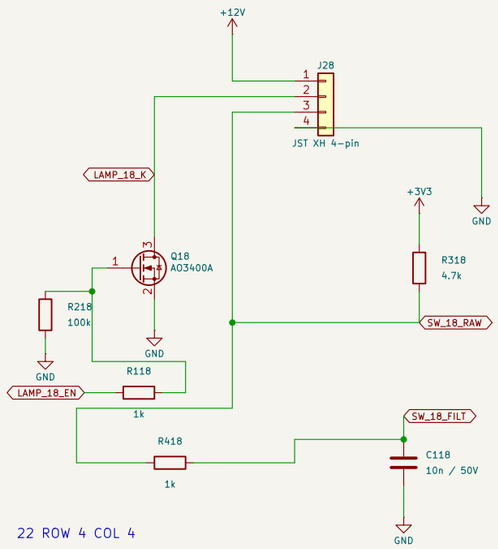
BUTTON HARNESS | bank 3 | four wires per illuminated button

Jx: 1=+12V lamp, 2=switched lamp cathode, 3=switch NO, 4=switch COM/GND.
Adafruit 1194 includes an LED resistor for operation up to 12 V. No extra series resistor fitted.
EG STARTS menu lamps specified as 12 V; current / tab dimensions need confirmation.
Use normally-open contacts; leave any NC switch terminal insulated and unused.

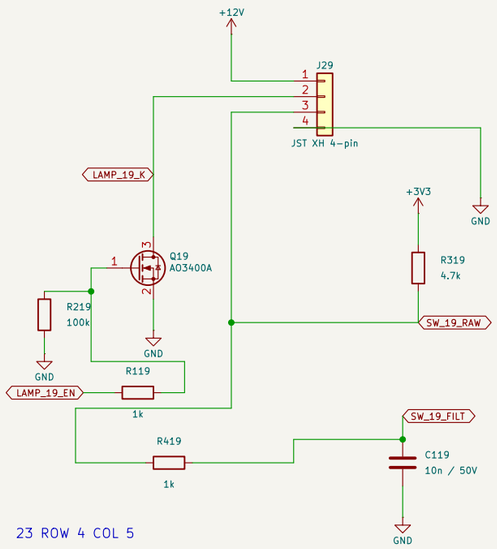
17 ROW 3 COL 5



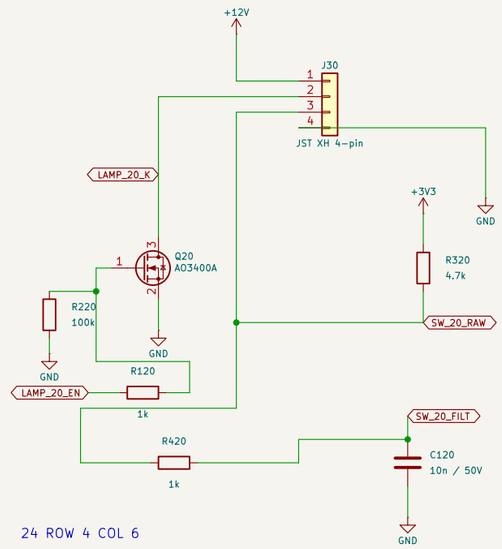
18 ROW 3 COL 6



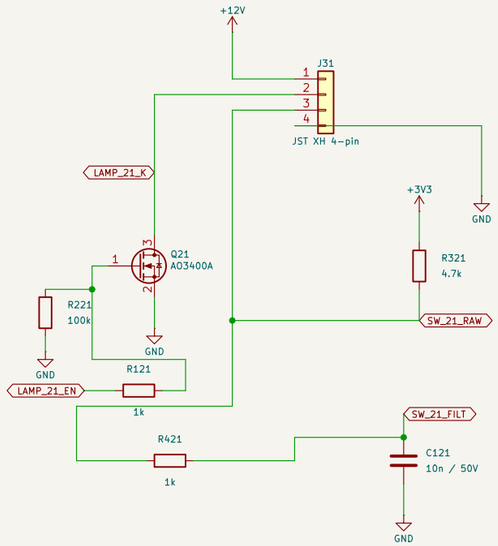
19 ROW 4 COL 1



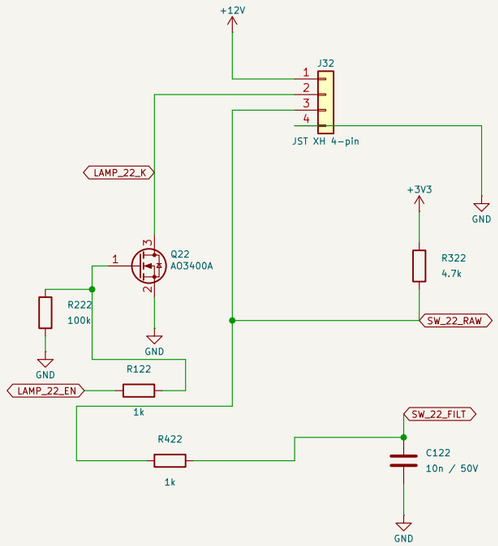
20 ROW 4 COL 2



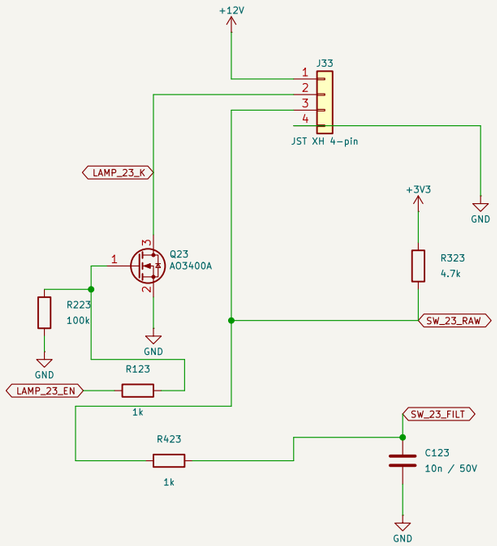
21 ROW 4 COL 3



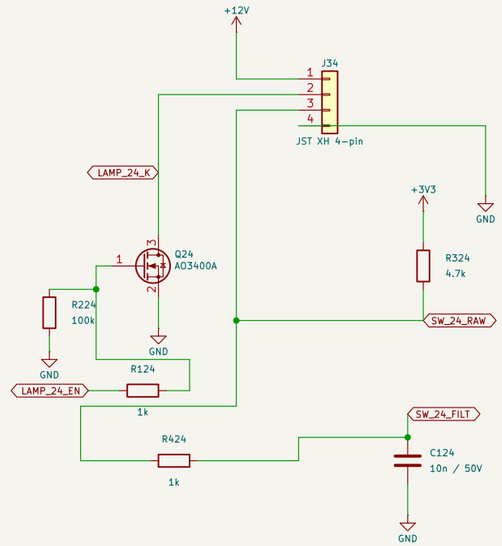
22 ROW 4 COL 4



23 ROW 4 COL 5



24 ROW 4 COL 6



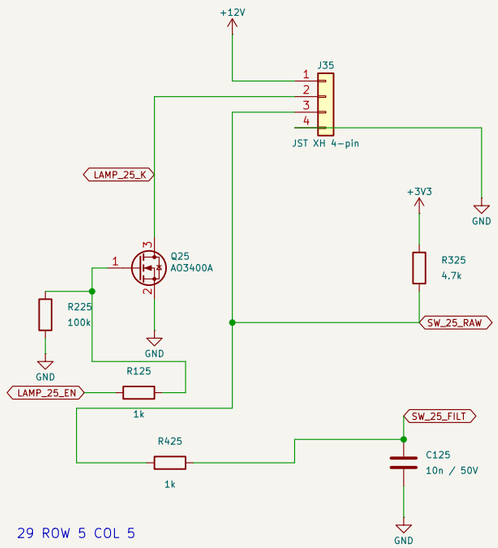
100 mA per lamp is a DESIGN ALLOCATION, not a measured lamp current. 32 lamps = 3.2 A.
XH cable mating parts: XHP-4 + 4 x SXH-001T-P0.6 per button. Accepts 22-28 AWG; strain-relieve.
Adafruit 1194: 3835 pigtails fit switch tabs; 3838 fit lamp tabs. See BOM for pack quantities.
Replace pigtail 2-pin ends with genuine XH contacts / 4-pin housing. Menu-button tab sizes UNVERIFIED.

UW-Madison IEEE		
Sheet: /Button circuits 3/		
File: button_bank_3.kicad_sch		
Title: Button harness / channels 17..24		
Size: A3	Date: 2026-11-05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 5/8

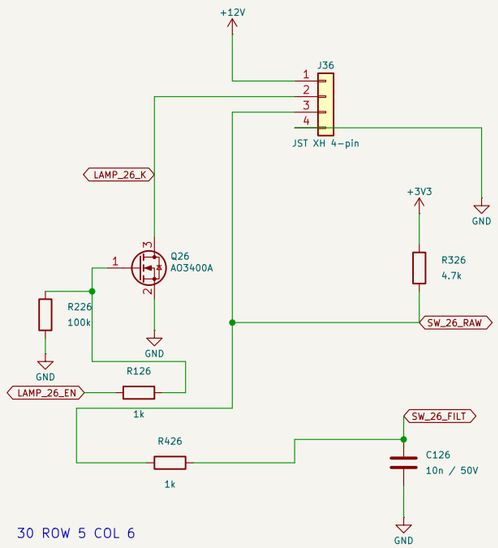
BUTTON HARNESS | bank 4 | four wires per illuminated button

Jx: 1=+12V lamp, 2=switched lamp cathode, 3=switch NO, 4=switch COM/GND.
Adafruit 1194 includes an LED resistor for operation up to 12 V. No extra series resistor fitted.
EG STARTS menu lamps specified as 12 V; current / tab dimensions need confirmation.
Use normally-open contacts; leave any NC switch terminal insulated and unused.

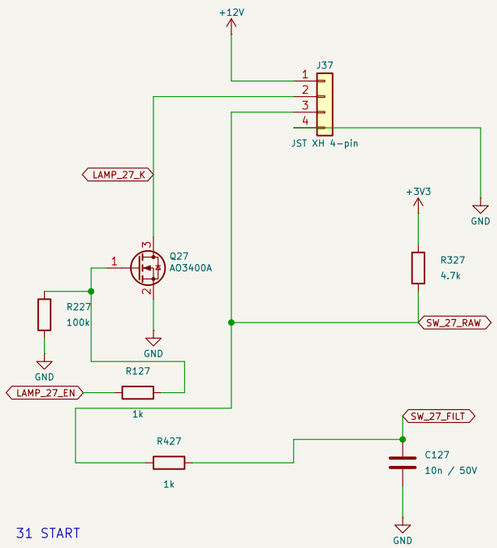
25 ROW 5 COL 1



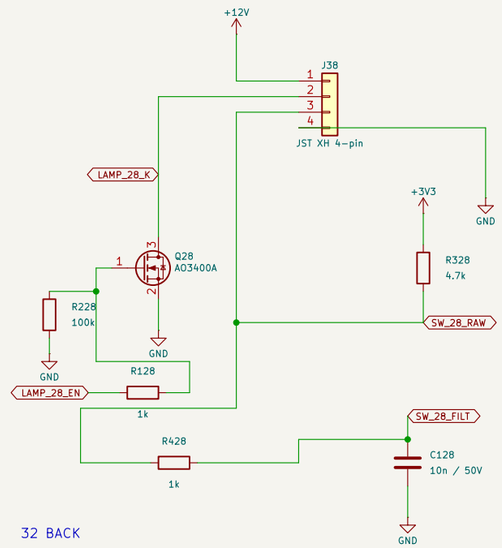
26 ROW 5 COL 2



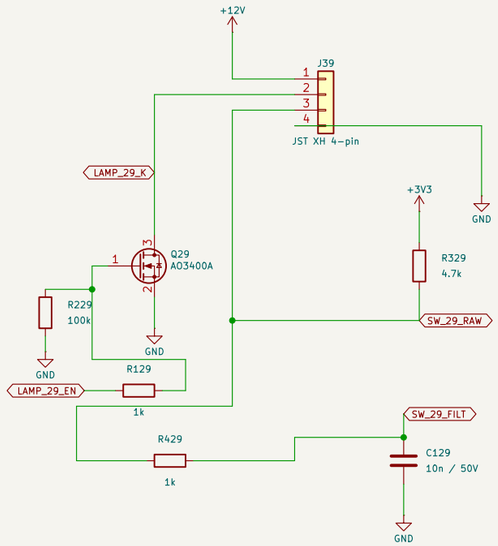
27 ROW 5 COL 3



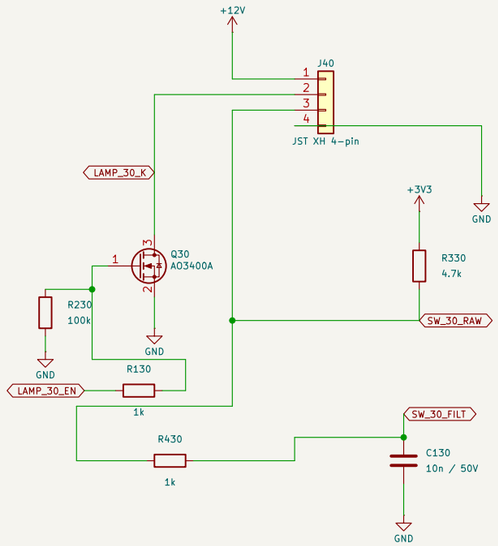
28 ROW 5 COL 4



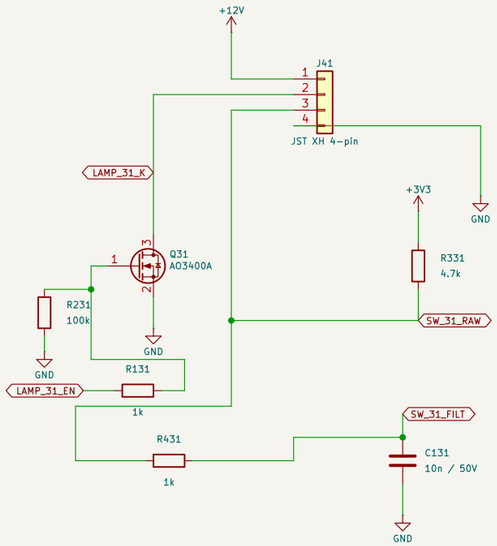
29 ROW 5 COL 5



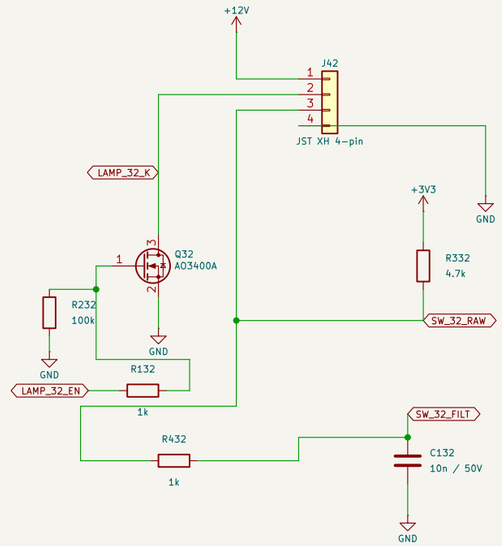
30 ROW 5 COL 6



31 START



32 BACK

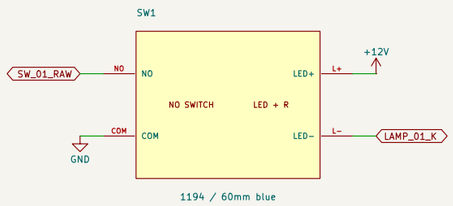


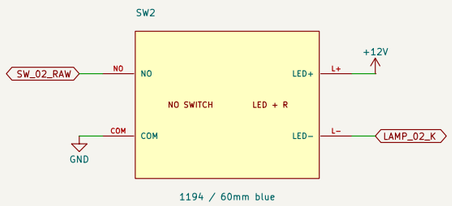
100 mA per lamp is a DESIGN ALLOCATION, not a measured lamp current. 32 lamps = 3.2 A.
XH cable mating parts: XHP-4 + 4 x SXH-001T-P0.6 per button. Accepts 22-28 AWG; strain-relieve.
Adafruit 1194: 3835 pigtails fit switch tabs; 3838 fit lamp tabs. See BOM for pack quantities.
Replace pigtail 2-pin ends with genuine XH contacts / 4-pin housing. Menu-button tab sizes UNVERIFIED.

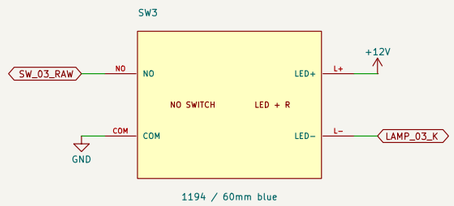
UW-Madison IEEE		
Sheet: /Button circuits 4/		
File: button_bank_4.kicad_sch		
Title: Button harness / channels 25..32		
Size: A3	Date: 2026-11-05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 6/8

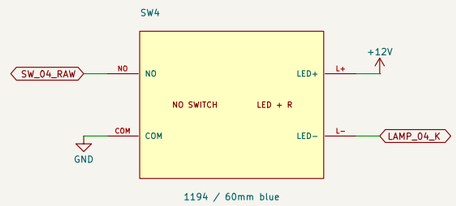
OFF-BOARD BUTTONS | parts mounted in cabinet, not on PCB

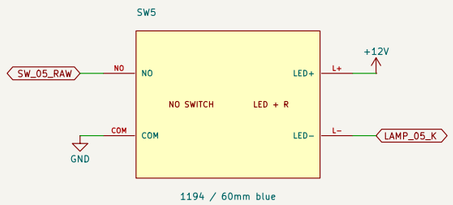
Functional pin names identify the wires; they are NOT manufacturer terminal numbers.
Microswitch COM/NO closes when pressed. Lamp has two separate terminals; observe LED polarity.
The light and switch circuits must remain separate. Insulate unused NC contacts, if present.
Adafruit 1194 includes a series LED resistor. Verify EG STARTS 12 V lamp module and current before use.

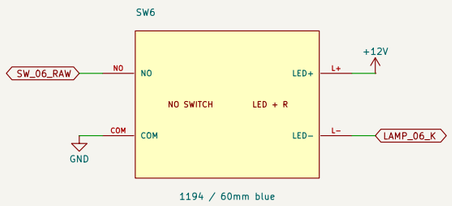


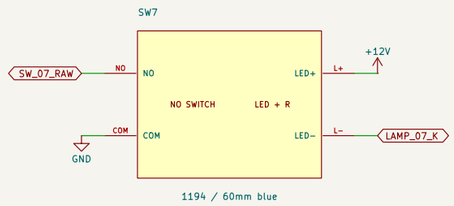


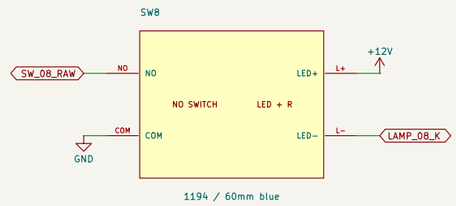


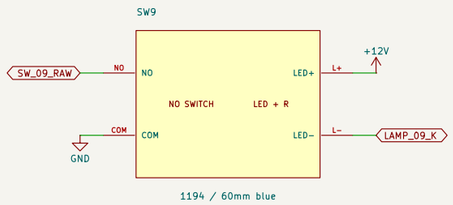


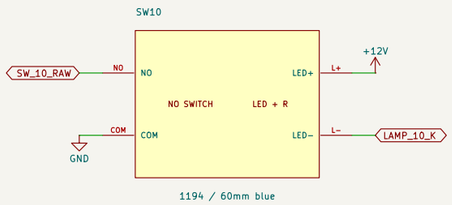


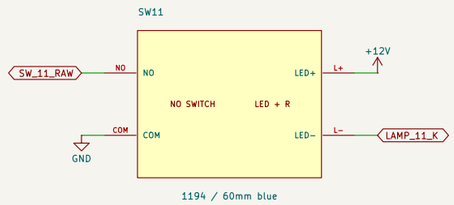


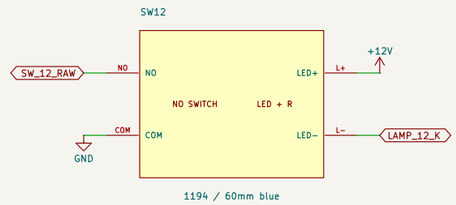


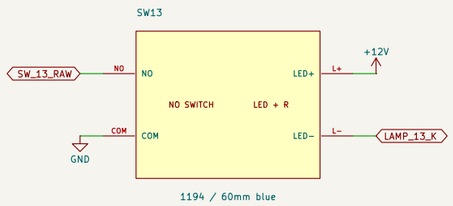


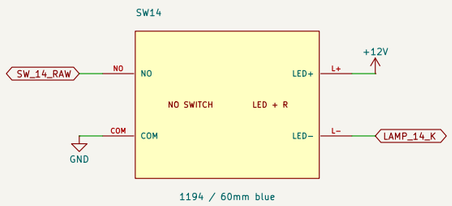


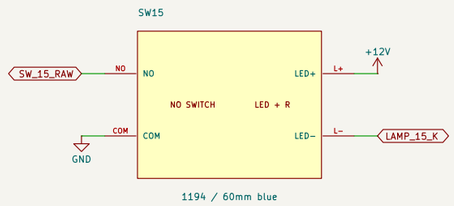


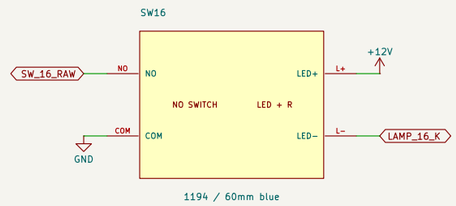












PCB harnesses are J11..J42. SW1..SW30 playfield; SW31 START; SW32 BACK.
No PCB footprints are assigned to cabinet buttons. Board-side headers and mating housings are specified in BOM.

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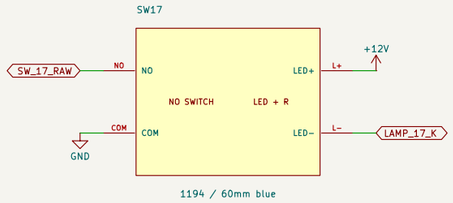
Sheet: /Panel-mounted buttons 1/
File: external_buttons_1.kicad_sch

Title: Off-board buttons 01..16

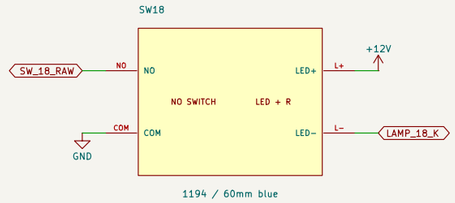
Size: A3	Date: 2026-11-05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 7/8

OFF-BOARD BUTTONS | parts mounted in cabinet, not on PCB

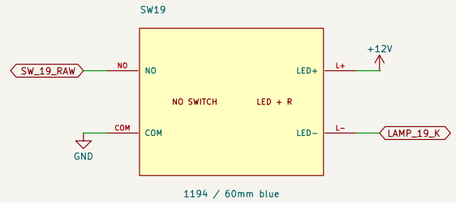
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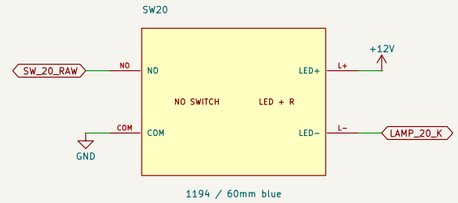
1194 / 60mm blue



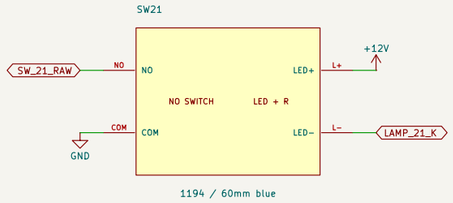
1194 / 60mm blue



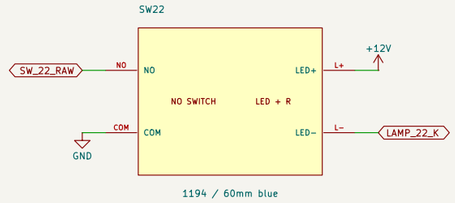
1194 / 60mm blue



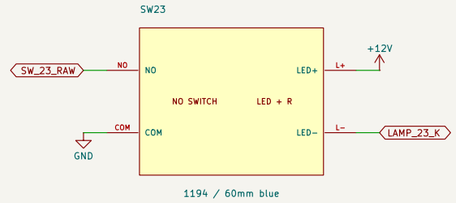
1194 / 60mm blue



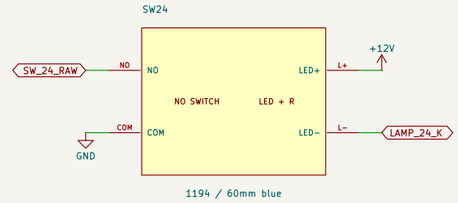
1194 / 60mm blue



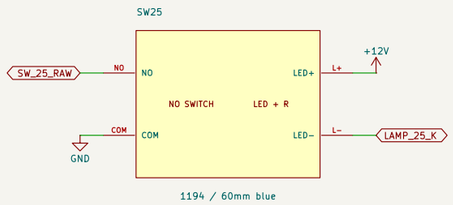
1194 / 60mm blue



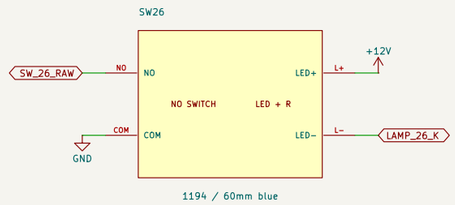
1194 / 60mm blue



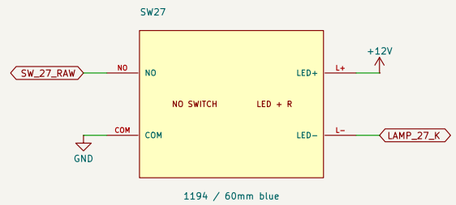
1194 / 60mm blue



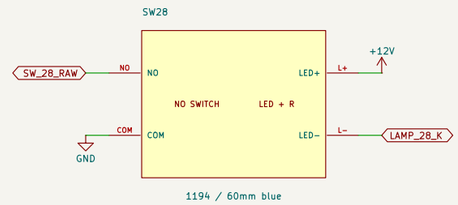
1194 / 60mm blue



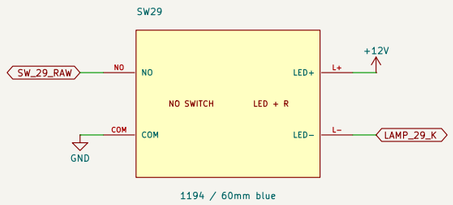
1194 / 60mm blue



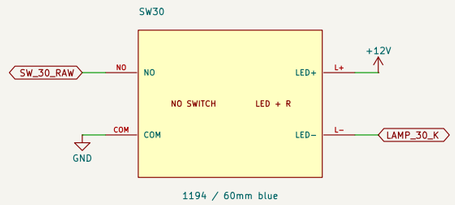
1194 / 60mm blue



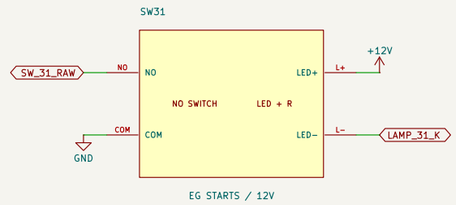
1194 / 60mm blue



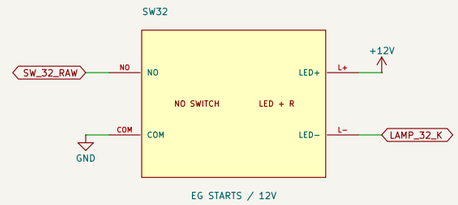
1194 / 60mm blue



1194 / 60mm blue



EG STARTS / 12V



EG STARTS / 12V

PCB harnesses are J11..J42. SW1..SW30 playfield; SW31 START; SW32 BACK.
No PCB footprints are assigned to cabinet buttons. Board-side headers and mating housings are specified in BOM.

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Sheet: /Panel-mounted buttons 2/
File: external_buttons_2.kicad_sch

Title: Off-board buttons 17..32

Size: A3	Date: 2026-11-05	**Rev: Schematic review**
KiCad E.D.A. 10.0.6		Id: 8/8